

# COMMUTATOR LENGTH OF ABELIAN-BY-NILPOTENT GROUPS

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**Abstract.** Let  $G$  be a group and  $G' = [G, G]$  be its commutator subgroup. Denote by  $c(G)$  the minimal number such that every element of  $G'$  can be expressed as a product of at most  $c(G)$  commutators. The exact values of  $c(G)$  are computed when  $G$  is a free nilpotent group or a free abelian-by-nilpotent group. If  $G$  is a free nilpotent group of rank  $n \geq 2$  and class  $c \geq 2$ ,  $c(G) = [n/2]$  if  $c = 2$  and  $c(G) = n$  if  $c > 2$ . If  $G$  is a free abelian-by-nilpotent group of rank  $n \geq 2$  then  $c(G) = n$ .

**1. Introduction.** For an element  $g$  in the derived subgroup  $G' = [G, G]$  of a group  $G$  we write  $c(g)$  to denote the least integer such that  $g$  can be written as a product of  $c(g)$  commutators and we put

$$c(G) = \sup\{c(g); g \in G'\}.$$

Let  $F_{n,t} = \langle x_1, \dots, x_n \rangle$  and  $M_{n,t} = \langle x_1, \dots, x_n \rangle$  be respectively the free nilpotent group of rank  $n$  and class  $t$  and the free metabelian nilpotent group of rank  $n$  and class  $t$ . P. W. Stroud, in his Ph.D. thesis [3] in 1966, proved that for all  $t$ , every element of the commutator subgroup  $F'_{n,t}$  can be expressed as a product of  $n$  commutators. In 1985 H. Allambergenov and V. A. Romankov [1] proved that  $c(M_{n,t})$  is precisely  $n$  provided  $n \geq 2$ ,  $t \geq 4$  or  $n \geq 3$ ,  $t \geq 3$ . They did this by producing an element  $d_n$  in  $\gamma_t(M_{n,t})$  that cannot be written as a product of fewer than  $n$  commutators. For the case  $n = 2$ ,  $c = 3$  they proved that every element of  $\gamma_3(M_{2,3})$  is a commutator, and claimed that  $c(M_{2,3})$  is one. We will show that the element  $[x_1, x_2]^2$  cannot be written as a commutator in the group  $M_{2,3} = F_{2,3} = \langle x_1, x_2 \rangle$ . This is done in Theorem 1. Thus  $c(F_{2,3}) = 2$  and  $c(F_{n,t}) = c(M_{n,t}) = n$ , for all  $n \geq 2$  and  $t \geq 3$ .

In [2] C. Bavard and G. Meigniez considered the same problem for the  $n$ -generator free metabelian group  $M_n$ . They show that the minimum number  $c(M_n)$  of commutators required to express an arbitrary element of the derived subgroup  $M'_n$  satisfies the inequality

$$[n/2] \leq c(M_n) \leq n,$$

where  $[n/2]$  is the greatest integer part of  $n/2$ . Since  $F_{n,3}$  groups are metabelian, the result of Allambergenov and Romankov [1] shows that  $c(M_n) \geq n$ , for  $n \geq 3$ , and Theorem 1 of this paper deals with the remaining case  $n = 2$ . We have  $c(M_n) = n$ , for all  $n \geq 2$ . Finally, we extend results in [1] and [2] to the larger class of abelian-by-nilpotent groups and show in Theorem 2 that  $c(G) = n$  if  $G$  is a (non-abelian) free abelian-by-nilpotent group of rank  $n$ .

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The main results of this paper are as follows.

**THEOREM 1.** *Let  $F_{2,3} = \langle x_1, x_2 \rangle$  be the free nilpotent group of class 3 on free generators  $x_1, x_2$ . Then  $c(F_{2,3}) = 2$ .*

**THEOREM 2.** *Let  $G = \langle x_1, \dots, x_n \rangle$  be a non-abelian free abelian-by-nilpotent group freely generated by  $x_1, \dots, x_n$ . Then  $c(G) = n$ . If  $A$  is an abelian normal subgroup of  $G$  and  $G/A$  is nilpotent, then every element of  $G'$  is a product of  $n$  commutators  $[x_1, g_1]^{a_1} [x_2, g_2]^{a_2} \dots [x_n, g_n]^{a_n}$ , for suitable  $g_1, \dots, g_n$  in  $G$  and  $a_1, \dots, a_n$  in  $A$ .*

If  $F_{n,2} = \langle x_1, \dots, x_n \rangle$  is the free nilpotent group of class two, then  $c(F_{n,2}) = n/2$ , if  $n$  is even and  $(n-1)/2$ , if  $n$  is odd. This result appears in [1].

We know of no example of a finite group  $G$  of rank  $n$  where  $c(G) > n$ . Nor do we know of any example of a solvable group  $G$  of rank  $n$  where  $c(G) > n$ .

**4. Proofs.** We begin by establishing a technical result required in the proof of Theorem 1.

**LEMMA 1.** *The following system of three equations in variables  $s_1, s_2, r_1, r_2, \alpha$  and  $\beta$  has no integer solution:*

$$r_2 s_1 - r_1 s_2 = 2, \quad (1)$$

$$\frac{s_1 r_2 (r_2 - 1)}{2} - \frac{r_1 s_2 (s_2 - 1)}{2} + r_2 s_2 (s_1 - r_1) - \alpha r_2 + \beta s_2 = 0, \quad (2)$$

$$\frac{r_2 s_2 (s_1 - 1)}{2} - \frac{r_1 s_2 (r_1 - 1)}{2} - \alpha r_1 + \beta s_1 = 0. \quad (3)$$

*Proof.* Put  $c_1 = \alpha r_2 - \beta s_2$ ,  $c_2 = \alpha r_1 - \beta s_1$ . Then

$$\alpha = \frac{\begin{vmatrix} c_1 & -s_2 \\ c_2 & -s_1 \end{vmatrix}}{\begin{vmatrix} r_2 & -s_2 \\ r_1 & -s_1 \end{vmatrix}} = \frac{s_1 c_1 - s_2 c_2}{2}, \quad \beta = \frac{\begin{vmatrix} r_2 & c_1 \\ r_1 & c_2 \end{vmatrix}}{-2} = \frac{r_1 c_1 - r_2 c_2}{2}.$$

Hence we need  $s_1 c_1 - s_2 c_2$  and  $r_1 c_1 - r_2 c_2$  to be even:

$$\begin{aligned} c_1 &= \frac{s_1 r_2 (r_2 - 1)}{2} - \frac{r_1 s_2 (s_2 - 1)}{2} + r_2 s_2 (s_1 - r_1), \\ c_2 &= \frac{r_2 s_1 (s_1 - 1)}{2} - \frac{r_1 s_2 (r_1 - 1)}{2}. \end{aligned}$$

Hence we have

$$\begin{aligned} 2c_1 &= s_1 r_2^2 - s_1 r_2 - r_1 s_2^2 + r_1 s_2 + 2s_1 r_2 s_2 - 2r_1 r_2 s_2, \\ &= r_2 (s_1 r_2 - r_1 s_2) - (s_1 r_2 - r_1 s_2) - s_2 (r_1 s_2 - s_1 r_2) - r_1 r_2 s_2 + s_1 r_2 s_2, \\ &= -2 + 2(r_2 + s_2) - r_2 s_2 (r_1 - s_1). \end{aligned}$$

Also we have

$$2c_2 = s_1^2 r_2 - s_1 r_2 - r_1^2 s_2 + r_1 s_2 = -2 + s_1^2 r_2 - r_1^2 s_2.$$

Hence we need to satisfy the following conditions:

$$r_2 s_1 - r_1 s_2 = 2, \quad (1)$$

$$2c_1 = -2 + 2(r_2 + s_2) - r_2 s_2(r_1 - s_1), \quad (4)$$

$$2c_2 = -2 + s_1^2 r_2 - r_1^2 s_2, \quad (5)$$

$$s_1 c_1 + s_2 c_2 \equiv r_1 c_1 + r_2 c_2 \equiv 0 \pmod{2}. \quad (6)$$

*Case 1.*  $r_1 s_2 = 2k$ , for some integer  $k$ . Then

$$c_1 = -1 + (r_2 + s_2) - k r_2 + (1 + k) s_2$$

$$c_2 = -1 + (1 + k) s_1 - k r_1.$$

Further

$$0 \equiv s_1 c_1 + s_2 c_2 \equiv s_1 + s_1 s_2 + s_2 \pmod{2} \quad (7)$$

and

$$0 \equiv r_1 c_1 + r_2 c_2 \equiv r_1 + r_1 r_2 + r_2 \pmod{2}. \quad (8)$$

From (7) and (8), it follows that  $r_1, r_2, s_1$  and  $s_2$  are all even. But then  $r_1 s_2 - r_2 s_1$  is divisible by 4, contradicting (1).

*Case 2.*  $r_1 s_2$  is odd. It follows from (1) that  $r_1, r_2, s_1, s_2$  are all odd.

$$\begin{aligned} s_1 c_1 + s_2 c_2 &= \frac{1}{2} s_1^2 r_2 (r_2 - 1) - \frac{1}{2} s_1 r_1 s_2 (s_2 - 1) + s_1 s_2 r_2 (s_1 - r_1) \\ &\quad + \frac{1}{2} s_2 r_2 s_1 (s_1 - 1) - \frac{1}{2} s_2^2 r_1 (r_1 - 1) \\ &= \frac{1}{2} (s_1 r_2 - s_2 r_1) (s_1 r_2 + s_2 r_1) \\ &\quad - \frac{1}{2} s_1 (s_1 r_2 - r_1 s_2) - \frac{1}{2} s_1 s_2 (r_1 s_2 - r_2 s_1) \\ &\quad - \frac{1}{2} s_2 (r_2 s_1 - s_2 r_1) + s_1 s_2 r_2 (s_1 - r_1) \\ &= (s_1 r_2 + s_2 r_1) - s_1 + s_1 s_2 - s_2 + s_1 s_2 r_2 (s_1 - r_1), \end{aligned}$$

which is odd and contradicts (7).

We shall use the following well known identities for groups which are nilpotent of class 3.

LEMMA 2. Let  $G = \langle x, y \rangle$  be nilpotent of class 3. Then, for all integers  $r, s$  the following hold:

$$\begin{aligned} [x^r, y] &= [x, y]^r [x, y, x]^{r(r-1)/2}, \\ [x^r, y^s] &= [x, y]^{rs} [x, y, x]^{rs(r-1)/2} [x, y, y]^{rs(s-1)/2}. \end{aligned}$$

*Proofs of Theorem 1.* Let  $h, g$  be any two elements of  $F_{2,3} \setminus \gamma_3(F_{2,3})$ . We study the form of the element  $[h, g]$ . Since  $\gamma_3(F_{2,3})$  lies in the center of  $F_{2,3}$  we may express  $h$  as

$x_1^r x_2^s [x_2, x_1]^\beta$  and  $g$  as  $x_1^{s_1} x_2^{s_2} [x_2, x_1]^\alpha$ . Put  $z = [x_2, x_1]$ ,  $y_1 = z^\beta$  and  $y_2 = z^\alpha$ . Then

$$\begin{aligned} [h, g] &= [x_1^r x_2^s, x_1^{s_1} x_2^{s_2}] [x_1^r x_2^s, y_2] [y_1, x_1^{s_1} x_2^{s_2}] \\ &= [x_1^r x_2^s, x_2^{s_2}] [x_1^r x_2^s, x_1^{s_1}] [x_1^r x_2^s, x_2^{s_2}] [x_1^r, y_2] [x_2^s, y_2] [y_1, x_2^{s_2}] [y_1, x_1^{s_1}] \\ &= [x_1^r, x_2^{s_2}] [x_1^r, x_2^s, x_2^{s_2}] [x_2^s, x_1^{s_1}] [x_2^s, x_1^{s_1}, x_2^{s_2}] [x_1, z]^{\alpha r_1} \\ &\quad \times [x_2, z]^{\alpha r_2} [z, x_2]^{\beta s_2} [z, x_1]^{\beta s_1} \\ &= [x_1, x_2]^{r_1 s_2} [x_1, x_2, x_1]^{s_2 r_1 (r_1 - 1)/2} [x_1, x_2, x_2]^{r_1 s_2 (s_2 - 1)/2} [x_1, x_2, x_2]^{r_1 r_2 s_2} \\ &\quad \times [x_2, x_1]^{r_2 s_1} [x_2, x_1, x_2]^{s_1 r_2 (r_2 - 1)/2} [x_1, x_1, x_1]^{r_2 s_1 (s_1 - 1)/2} [x_2, x_1, x_2]^{r_2 s_1 s_2} \\ &\quad \times [x_2, x_1, x_1]^{-\alpha r_1 + \beta s_1} [x_1, x_1, x_2]^{-\alpha r_2 + \beta s_2} \\ &= [x_2, x_1]^\lambda [x_2, x_1, x_2]^\mu [x_2, x_1, x_1]^\nu, \end{aligned}$$

where

$$\begin{aligned} \lambda &= r_2 s_1 - r_1 s_2, \\ \mu &= \frac{s_1 r_2 (r_2 - 1)}{2} - \frac{r_1 s_2 (s_2 - 1)}{2} - r_1 r_2 s_2 + r_2 s_1 s_2 + \beta s_2 - \alpha r_2, \\ \nu &= \frac{r_2 s_1 (s_1 - 1)}{2} - \frac{s_2 r_1 (r_1 - 1)}{2} + \beta s_1 - \alpha r_1. \end{aligned}$$

Since  $[x_2, x_1]$ ,  $[x_2, x_1, x_2]$  and  $[x_2, x_1, x_1]$  are the basic commutators and the group under consideration is the free nilpotent class 3 group, it follows that if  $[h, g] = [x_2, x_1]^2$  then  $\lambda = 2$ ,  $\mu = \nu = 0$ . But, by Lemma 1, there are no integers  $\alpha, \beta, r_1, s_1, r_2, s_2$  for this set of equations to hold and we conclude that  $c(F_{2,3}) \geq 2$ . Since  $c(F_{2,3}) \leq 2$  by [1] or [2], we obtain the equality.

The proof of Theorem 2 makes use of the following two results. The first is elementary; the second is a result of Peter Stroud [4]. We shall include the proofs since Stroud's result never got published, except in his Ph.D. thesis, due to his untimely death. In the case of a finite group  $G$ , Brian Hartley [3] has given a bound for  $c(G)$  in terms of the Fitting length of  $G$ . His proof incorporates Stroud's proof given below.

LEMMA 3. *Let  $A$  be a normal subgroup of  $G = \langle x_1, \dots, x_n \rangle$ . If  $A$  is abelian or  $A$  lies in the second center  $\zeta_2(G)$  of  $G$ , then every element of  $[G, A]$  has the form  $\prod_{i=1}^n [x_i, a_i]$ , where  $a_i \in A$ .*

*Proof.* Consider  $[g, d]$ , where  $d \in A$  and  $g = x_{i_1}^{\varepsilon_1} \dots x_{i_r}^{\varepsilon_r}$ , where  $\varepsilon_i \in \{1, -1\}$ . Write  $x_{i_1} = x$ ,  $\varepsilon_1 = \varepsilon$  and  $g = x^\varepsilon y$ . Then  $[g, d] = [x^\varepsilon y, d] = [x, d^\varepsilon][y, d]$ , if  $A \leq \zeta_2(G)$ , and  $[g, d] = [x^\varepsilon, d][x^\varepsilon, d, y][y, d] = [x^\varepsilon, d][y, d[d, x^\varepsilon]]$ , if  $A$  is abelian. If  $\varepsilon = -1$ , then use  $[x^{-1}, d] = [x, xd^{-1}x^{-1}]$ .

Iterate the process  $r$  times to obtain  $[g, d] = \prod_{j=1}^r [x_{ij}, d_j]$  with  $d_j \in A$ . Finally use the identity  $[x, d_1][x, d_2] = [x, d_1 d_2]$  to see that every  $\prod [g_i, d_i]$ ,  $d_i \in A$  has the form  $\prod_{i=1}^n [x_i, a_i]$ , where  $a_i \in A$ .

LEMMA (P. Stroud). *Let  $G = \langle x_1, \dots, x_n \rangle$  be a nilpotent group. Then every element of the commutator subgroup  $G'$  is a product of  $n$  commutators  $[x_1, g_1] \dots [x_n, g_n]$ , for suitable  $g_i$  in  $G$ .*

*Proof.* Use induction on the nilpotency class of  $G$ . If  $G$  is abelian, then  $G' = 1$  and the result is clear. Next let  $G \in \mathcal{N}_{r+1}$ , nilpotent of class  $r+1$ , and assume the result for groups in the class  $\mathcal{N}_r$ . Let  $\Gamma = \gamma_{r+1}(G) = [\gamma_r(G), G]$ . Then an element  $g$  of  $G'$  has the form  $g = [x_1, h_1] \dots [x_n, h_n]d$ , for some  $d \in \Gamma$ . By Lemma 3, we have

$$g = [x_1, h_1] \dots [x_n, h_n][x_1, a_1] \dots [x_n, a_n] = \prod_{i=1}^n [x_i, h_i a_i].$$

*Proof of Theorem 2.* By hypothesis, there exists a normal abelian subgroup  $A$  of  $G$  such that  $G/A$  is nilpotent. By Lemma 3,  $[A, G] = \{[x_1, a_1] \dots [x_n, a_n]; a_i \in A\}$  and, since  $G/[A, G]$  is nilpotent, using Stroud's result, every element  $g \in G'$  has the form

$$\begin{aligned} g &= \left( \prod_{i=1}^n [x_i, g_i] \right) \left( \prod_{i=1}^n [x_i, a_i] \right), \text{ with } a_i \in A, \\ &= \prod_{i=1}^n ([x_i, a_i][x_i, g_i]^{d_i}), \text{ for suitable } d_i \in A. \end{aligned}$$

$$\text{Now } [x_i, g_i a_i] = [x_i, a_i][x_i, g_i]^{a_i} = ([x_i, a_i][x_i, g_i])^{a_i}.$$

Thus  $[x_i, a_i][x_i, g_i] = [x_i, g_i a_i]^{a_i^{-1}}$  and  $g = \prod_{i=1}^n [x_i, g_i a_i]^{d_i a_i^{-1}}$ , with  $d_i a_i^{-1} \in A$ . Thus  $c(G) \leq n$  and every element of  $G'$  has the required form. Since  $G$  is free abelian-by-nilpotent and non-abelian, the free metabelian group on  $n$ -generators is a quotient of  $G$  and hence so is the free nilpotent-class-three group on  $n$  generators. By Theorem 1 for the case when  $n = 2$  and by [1] for  $n > 2$  we have  $c(G) \geq n$ . This shows that  $c(G) = n$  and completes the proof.

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